

61. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x}{\sqrt{1+9x^2}}$. If the composition of

$$f, \underbrace{(f \circ f \circ f \circ \dots \circ f)}_{10 \text{ times}}(x) = \frac{2^{10}x}{\sqrt{1+9\alpha x^2}}, \text{ then the value of } \sqrt{3\alpha+1} \text{ is equal to } \underline{\hspace{2cm}}.$$

विचार कीजिए फलन $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{2x}{\sqrt{1+9x^2}}$ द्वारा परिभाषित है। यदि f का संयोजन

$$f, \underbrace{(f \circ f \circ f \circ \dots \circ f)}_{10 \text{ times}}(x) = \frac{2^{10}x}{\sqrt{1+9\alpha x^2}} \text{ है, तो } \sqrt{3\alpha+1} \text{ का मान बराबर है } \underline{\hspace{2cm}}$$

Ans. 1024

Sol. $f(x) = \frac{2x}{\sqrt{1+9x^2}}$

$$(f \circ f)(x) = \frac{2f(x)}{\sqrt{1+9(f(x))^2}} = \frac{\frac{4x}{\sqrt{1+9x^2}}}{\sqrt{1+9 \times \frac{4x^2}{1+9x^2}}} = \frac{4x}{\sqrt{1+45x^2}}$$

$$(f \circ f \circ f)(x) = \frac{4 \times \frac{2x}{\sqrt{1+9x^2}}}{\sqrt{1+45 \times \frac{4x^2}{1+9x^2}}} = \frac{8x}{\sqrt{1+21 \times 9x^2}}$$

$$(f \circ f \circ f \circ f)(x) = \frac{16x}{\sqrt{1+85 \times 9x^2}}$$

$\Rightarrow \alpha$ is 10th term of 1, 5, 21, 85,

$$\alpha \text{ is 10}^{\text{th}} \text{ term of } \frac{(2^1)^2-1}{3}, \frac{(2^2)^2-1}{3}, \frac{(2^3)^2-1}{3}, \frac{(2^4)^2-1}{3}, \dots$$

$$\Rightarrow \alpha = \frac{(2^{10})^2-1}{3}$$

$$\Rightarrow \sqrt{3\alpha+1} = 2^{10} = 1024$$

62. If $S = \{a \in \mathbb{R} : |2a-1| = 3[a] + 2\{a\}\}$, where $[t]$ denotes the greatest integer less than or equal to t and $\{t\}$ represents

the fractional part t , then $72 \sum_{a \in S} a$ is equal to _____.

यदि $S = \{a \in \mathbb{R} : |2a - 1| = 3[a] + 2\{a\}\}$ है, जहाँ $[t]$ महत्तम पूर्णांक फलन है तथा $\{t\}$, t के भिन्नात्मक भाग को दर्शाता है, तो

$72 \sum_{a \in S} a$ बराबर है _____

Ans. 18

Sol. $|2a - 1| = 3[a] + 2\{a\}$

Case-I : $a \geq \frac{1}{2}$

$$2a - 1 = [a] + 2a \quad (\because a = [a] + \{a\})$$

$$[a] = -1$$

$$a \in [-1, 0) \text{ but } a \geq \frac{1}{2}, \Rightarrow \text{No value of } a$$

Case-II: $a < \frac{1}{2}$

$$1 - 2a = [a] + 2a$$

$$1 = [a] + 4a$$

$$1 = 5[a] + 4\{a\}$$

$$4\{a\} = 1 - 5[a]$$

$$0 \leq 4\{a\} < 4$$

$$0 \leq 1 - 5[a] < 4$$

$$\frac{1}{3} \geq [a] \geq -\frac{3}{5} \Rightarrow [a] = 0 \Rightarrow \{a\} = \frac{1}{4}$$

$$\Rightarrow a = \frac{1}{4}$$

$$\text{Now, } 72 \sum a = 72 \times \frac{1}{4} = 18$$

63. If a function f satisfies $f(m+n) = f(m) + f(n)$ for all $m, n \in \mathbb{N}$ and $f(1) = 1$, then the largest natural number λ such

that $\sum_{k=1}^{2022} f(\lambda + k) \leq (2022)^2$ is equal to _____.

यदि सभी $m, n \in \mathbb{N}$ के लिए, $f(m+n) = f(m) + f(n)$ है तथा $f(1) = 1$ है, के अधिकतम धन पूर्णांक λ जिसके लिए

$\sum_{k=1}^{2022} f(\lambda + k) \leq (2022)^2$ है, है _____

Ans. 1010

Sol. $f(m+n) = f(m) + f(n)$

$$\Rightarrow f(x) = kx$$

$$\Rightarrow f(1) = 1$$

$$\Rightarrow k = 1$$

$$f(x) = x$$

Now

$$\sum_{k=1}^{2022} f(\lambda + k) \leq (2022)^2$$

$$\Rightarrow \sum_{k=1}^{2022} (\lambda + k) \leq (2022)^2$$

$$\Rightarrow 2022\lambda + \frac{2022 \times 2023}{2} \leq (2022)^2$$

$$\Rightarrow \lambda \leq 2022 - \frac{2023}{2}$$

$$\Rightarrow \lambda \leq 1010.5$$

64. Let $A = \{(x, y) : 2x + 3y = 23, x, y \in \mathbb{N}\}$ and $B = \{x : (x, y) \in A\}$. Then the number of one-one functions from A to B is equal to _____.

माना $A = \{(x, y) : 2x + 3y = 23, x, y \in \mathbb{N}\}$ तथा $B = \{x : (x, y) \in A\}$ हैं। तो A से B के एकैकी फलनों की संख्या है _____.

Ans. 24

Sol. $2x + 3y = 23$

$$x = 1 \quad y = 7$$

$$x = 4 \quad y = 5$$

$$x = 7 \quad y = 3$$

$$x = 10 \quad y = 1$$

$$A \quad B$$

$$(1, 7) \quad 1$$

$$(4, 5) \quad 4$$

$$(7, 3) \quad 7$$

$$(10, 1) \quad 10$$

The number of one-one functions from A to B is equal to 4!

65. Let $A = \{1, 2, 3, \dots, 7\}$ and let $P(A)$ denote the power set of A. If the number of functions $f: A \rightarrow P(A)$ such that $a \in f(a), \forall a \in A$, is m^n , m and $n \in \mathbb{N}$ and m is least, then $m + n$ is equal to _____.

माना $A = \{1, 2, 3, \dots, 7\}$ है तथा माना समुच्चय A का घात (Power) समुच्चय $P(A)$ है। यदि फलनों $f: A \rightarrow P(A)$, जिनके लिए $a \in f(a), \forall a \in A$ है, की संख्या m^n , $m, n \in \mathbb{N}$ हैं तथा m न्यूनतम है, तो $m + n$ बराबर है _____।

Ans. 44

Sol. $n(P(A)) = 2^7$

$$a \in f(a)$$

$\Rightarrow f(a)$ will have (2^6) different subsets having a in them as choice.

$$\Rightarrow (2^6)^7 = 2^{42} \text{ function}$$

$$m^n = 2^{42}$$

$$\Rightarrow m + n = 44$$

66. Let $R = \{a, b, c, d, e\}$ and $S = \{1, 2, 3, 4\}$. Total number of onto functions $f: R \rightarrow S$ such that $f(a) \neq 1$, is equal to _____.

माना $R = \{a, b, c, d, e\}$ तथा $S = \{1, 2, 3, 4\}$ हैं। आच्छादक फलनों $f: R \rightarrow S$ जिनके लिए $f(a) \neq 1$ है, की कुल संख्या है :

Ans. 180

Sol. Total onto function

$$\frac{|5|}{|3|} \times |4| = 240$$

Now when $f(a) = 1$

$$|4| + \frac{|4|}{|2|} \times |3| = 24 + 36 = 60$$

$$\text{so required } f^n = 240 - 60 = 180$$

67. Let m and n be the numbers of real roots of the quadratic equations $x^2 - 12x + [x] + 31 = 0$ and $x^2 - 5[x + 2] - 4 = 0$ respectively, where $[x]$ denotes the greatest integer $\leq x$. Then $m^2 + mn + n^2$ is equal to _____.

माना समीकरणों $x^2 - 12x + [x] + 31 = 0$ तथा $x^2 - 5[x + 2] - 4 = 0$ के वास्तविक मूलों की संख्या क्रमशः m तथा n है, जहाँ $[x]$ महत्तम पूर्णांक $\leq x$ है। तो $m^2 + mn + n^2$ बराबर है :

Ans. 9

Sol. $x^2 - 12x + [x] + 31 = 0$

$$\underbrace{x^2 - 12x + 31}_{\geq -5} + [x] = 0$$

It could have its solution in $[5, 6]$ but it does not exist as at $x = 5$ and 6

$$\text{LHS} = 1$$

$$m = 0$$

$$x^2 - 5|x + 2| - 4 = 0$$

Case 1

$$x \geq -2$$

$$x^2 - 5(x + 2) - 4 = 0$$

$$x^2 - 5x - 14 = 0 \Rightarrow x = 7, -2$$

Case 2

$$x < -2$$

$$x^2 + 5x + 10 - 4 = 0$$

$$x = -2, -3$$

3 solution i.e., $x = -3, -2, 7$

$$n = 3$$

68. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 2, 3, 4, 5, 6\}$. Then the number of functions $f : A \rightarrow B$ satisfying $f(1) + f(2) = f(4) - 1$ is equal to _____.

माना $A = \{1, 2, 3, 4, 5\}$ तथा $B = \{1, 2, 3, 4, 5, 6\}$ हैं। तो $f(1) + f(2) = f(4) - 1$ को संतुष्ट करने वाले फलनों $f : A \rightarrow B$ की संख्या है _____

Ans. 360

Sol. $f(1) + f(2) + 1 = f(4)$

$$\Rightarrow f(1) + f(2) \leq 5$$

$$\text{Let } f(4) = 6, \text{ then } f(1) + f(2) = 5$$

So (f_1, f_2) may be $(1, 4), (4, 1), (2, 3), (3, 2)$

$$\text{Total ways} = 4 \cdot 1 \cdot 6 \cdot 6 = 144$$

$$\text{Let } f(4) = 5, \quad f(1) + f(2) = 4$$

so (f_1, f_2) may be $(1, 3), (3, 1), (2, 2)$

$$\text{Total ways} = 3 \cdot 1 \cdot 6 \cdot 6 = 108$$

$$\text{Let } f(4) = 4, \quad f(1) + f(2) = 3$$

so (f_1, f_2) may be $(1, 2), (2, 1)$

$$\text{Total ways} = 2 \cdot 1 \cdot 6 \cdot 6 = 72$$

$$\text{Let } f(4) = 3, \quad f(1) + f(2) = 2$$

Only way $(1, 1)$

$$\text{Total ways} = 1 \cdot 1 \cdot 36 = 36$$

69. Let $[\alpha]$ denote the greatest integer $\leq \alpha$. Then $[\sqrt{1}] + [\sqrt{2}] + [\sqrt{3}] + \dots + [\sqrt{120}]$ is equal to _____.

माना $[\alpha]$ महत्तम पूर्णांक $\leq \alpha$ है। तो $[\sqrt{1}] + [\sqrt{2}] + [\sqrt{3}] + \dots + [\sqrt{120}]$ बराबर है _____

Ans. 825

Sol. $S = [\sqrt{1}] + [\sqrt{2}] + [\sqrt{3}] + \dots + [\sqrt{120}]$

$$[\sqrt{1}] \rightarrow [\sqrt{3}] = 1 \times 3$$

$$[\sqrt{4}] \rightarrow [\sqrt{8}] = 2 \times 5$$

$$[\sqrt{9}] \rightarrow [\sqrt{15}] = 3 \times 7$$

$\vdots$

$$[\sqrt{100}] \rightarrow [\sqrt{120}] = 10 \times 21$$

$$S = 1 \times 3 + 2 \times 5 + 3 \times 7 + \dots + 10 \times 21$$

$$= \sum_{r=1}^{10} r(2r+1)$$

$$= 2 \sum_{r=1}^{10} r^2 + \sum_{r=1}^{10} r$$

$$= \frac{2 \times 10 \times 11 \times 21}{6} + \frac{10 \times 11}{2}$$

$$= 770 + 55$$

$$= 825$$

70. Let $f(x) = \sum_{k=1}^{10} k x^k, x \in \mathbb{R}$. If $2f(2) + f'(2) = 119(2)^n + 1$ then n is equal to _____.

माना $f(x) = \sum_{k=1}^{10} k x^k, x \in \mathbb{R}$ है। यदि $2f(2) + f'(2) = 119(2)^n + 1$ है, तो n बराबर है _____

Ans. 10

Sol. $f(x) = \sum_{k=1}^{10} k x^k$

$$f(x) = x + 2x^2 + \dots + 10x^{10}$$

$$f(x) \cdot x = x^2 + 2x^3 + \dots + 9x^{10} + 10x^{11}$$

$$f(x)(1-x) = x + x^2 + x^3 + \dots + x^{10} - 10x^{11}$$

$$f(x) = \frac{x(1-x^{10})}{(1-x)^2} - \frac{10x^{11}}{(1-x)}$$

$$f(x) = \frac{x - x^{11} - 10x^{11} + 10x^{12}}{(1-x)^2} \Rightarrow \frac{10x^{12} - 11x^{11} + x}{(1-x)^2}$$

Hence $2f(2) + f'(2) = 119 \cdot 2^{10} + 1$
 $\Rightarrow$ So, $n = 10$

Inverse Trigonometry Function

71. Considering the principal values of the inverse trigonometric functions, $\sin^{-1}\left(\frac{\sqrt{3}}{2}x + \frac{1}{2}\sqrt{1-x^2}\right)$, $-\frac{1}{2} < x < \frac{1}{\sqrt{2}}$, is equal to :

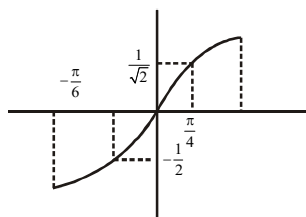
प्रतिलोम त्रिकोणमितीय फलनों के मुख्य मान लेने पर $\sin^{-1}\left(\frac{\sqrt{3}}{2}x + \frac{1}{2}\sqrt{1-x^2}\right)$, $-\frac{1}{2} < x < \frac{1}{\sqrt{2}}$ का मान होगा।

- (A) $\frac{5\pi}{6} - \sin^{-1} x$ (B) $\frac{\pi}{4} + \sin^{-1} x$ (C) $\frac{-5\pi}{6} - \sin^{-1} x$ (D) $\frac{\pi}{6} + \sin^{-1} x$

Ans. D

Sol. $\sin^{-1}\left(\frac{\sqrt{3}}{2}x + \frac{1}{2}\sqrt{1-x^2}\right)$; $-\frac{1}{2} < x < \frac{1}{\sqrt{2}}$

Let $x = \sin \theta$



$$-\frac{\pi}{6} < \theta < \frac{\pi}{4}$$

$$\sin^{-1}\left[\frac{\sqrt{3}}{2}\sin \theta + \frac{1}{2}\cos \theta\right]$$

$$\sin^{-1}\left[\cos \frac{\pi}{6}\sin \theta + \sin \frac{\pi}{6}\cos \theta\right]$$

$$\sin^{-1}\left(\sin\left(\theta + \frac{\pi}{6}\right)\right)$$

$$0 < \theta + \frac{\pi}{6} < \frac{\pi}{4} + \frac{\pi}{6}$$

$$0 < \theta + \frac{\pi}{6} < \frac{5\pi}{12}$$

$$\Rightarrow \theta + \frac{\pi}{6}$$

$$\Rightarrow \frac{\pi}{6} + \sin^{-1} x$$

72. Let the domains of the function $f(x) = \log_4 \log_3 \log_7 (8 - \log_2 (x^2 + 4x + 5))$ and $g(x) = \sin^{-1} \left(\frac{7x+10}{x-2} \right)$ be (α, β) and $[\gamma, \delta]$, respectively. Then $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ is equal to :

माना फलनों $f(x) = \log_4 \log_3 \log_7 (8 - \log_2 (x^2 + 4x + 5))$ और $g(x) = \sin^{-1} \left(\frac{7x+10}{x-2} \right)$ के प्रान्त क्रमशः (α, β) और

$[\gamma, \delta]$ हैं। तो $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ बराबर है :

(A) 13

(B) 15

(C) 14

(D) 16

Ans. B

Sol. $f(x) = \log_4 \log_3 \log_7 (8 - \log_2 (x^2 + 4x + 5))$

$$\Rightarrow x^2 + 4x + 5 > 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow 8 - \log_2 (x^2 + 4x + 5) > 0 \quad \dots(1)$$

$$\Rightarrow \log_7 (8 - \log_2 (x^2 + 4x + 5)) > 0$$

$$= 8 - \log_2 (x^2 + 4x + 5) > 1 \quad \dots(2)$$

$$\Rightarrow \log_3 \log_7 (8 - \log_2 (x^2 + 4x + 5)) > 0$$

$$= 8 - \log_2 (x^2 + 4x + 5) > 7 \quad \dots(3)$$

from (1), (2), (3)

$$\log_2 (x^2 + 4x + 5) < 1$$

$$x^2 + 4x + 5 < 2$$

$$x^2 + 4x + 3 < 0$$

$$(x+1)(x+3) < 0 \Rightarrow x \in (-3, -1)$$

$$g(x) = \sin^{-1} \left(\frac{7x+10}{x-2} \right)$$

$$-1 \leq \frac{7x+10}{x-2} \leq 1$$

$$\frac{7x+10}{x-2} + 1 \geq 0$$

&

$$\frac{7x+10}{x-2} - 1 \leq 0$$

$$\frac{8x+8}{x-2} \geq 0$$

$$\frac{6x+12}{x-2} \leq 0$$

$$x \in (-\infty, -1] \cup (2, \infty)$$

$$x \in [-2, 2)$$

$$D_g : [-2, -1]$$

$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 9 + 1 + 4 + 1 = 15$$

73. The sum of the infinite series $\cot^{-1}\left(\frac{7}{4}\right) + \cot^{-1}\left(\frac{19}{4}\right) + \cot^{-1}\left(\frac{39}{4}\right) + \cot^{-1}\left(\frac{67}{4}\right) + \dots$ is :

अनंत श्रेणी $\cot^{-1}\left(\frac{7}{4}\right) + \cot^{-1}\left(\frac{19}{4}\right) + \cot^{-1}\left(\frac{39}{4}\right) + \cot^{-1}\left(\frac{67}{4}\right) + \dots$ का योग है :

(A) $\frac{\pi}{2} - \cot^{-1}\left(\frac{1}{2}\right)$ (B) $\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{2}\right)$ (C) $\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{2}\right)$ (D) $\frac{\pi}{2} + \cot^{-1}\left(\frac{1}{2}\right)$

Ans. B

Sol. $\tan^{-1}\left(\frac{4}{7}\right) + \tan^{-1}\left(\frac{4}{19}\right) + \tan^{-1}\left(\frac{4}{39}\right) + \dots$

$$S = 7 + 19 + 39 + 67 + \dots + T_n$$

$$S = 7 + 19 + 39 + \dots + T_n$$

—

$$T_n = 7 + (12 + 20 + 28 + \dots (n-1) \text{ terms})$$

$$T_n = 7 + \frac{n-1}{2}(24 + (n-2)8)$$

$$T_n = 7 + (n-1)(4 + 4n)$$

$$T_n = 7 + 4(n^2 - 1) = 4n^2 + 3$$

$$\sum_{n=1}^{\infty} \tan^{-1}\left(\frac{4}{4n^2 + 3}\right) = \sum_{n=1}^{\infty} \tan^{-1}\left(\frac{1}{n^2 + \frac{3}{4}}\right)$$

$$\Rightarrow \sum_{n=1}^{\infty} \tan^{-1}\left(\frac{\left(n + \frac{1}{2}\right) - \left(n - \frac{1}{2}\right)}{1 + \left(n - \frac{1}{2}\right)\left(n + \frac{1}{2}\right)}\right)$$

$$\Rightarrow \sum_{n=1}^{\infty} \left(\tan^{-1} \left(n + \frac{1}{2} \right) - \tan^{-1} \left(n - \frac{1}{2} \right) \right)$$

$$\Rightarrow \frac{\pi}{2} - \tan^{-1} \frac{1}{2}$$

74. Using the principal values of the inverse trigonometric functions, the sum of the maximum and the minimum values of $16((\sec^{-1}x)^2 + (\operatorname{cosec}^{-1}x)^2)$ is :

प्रतिलोम त्रिकोणमितीय फलनों के मुख्य मानों का प्रयोग करते हुए $16((\sec^{-1}x)^2 + (\operatorname{cosec}^{-1}x)^2)$ के अधिकतम और न्यूनतम मानों का योग है :

- (A) $31\pi^2$ (B) $24\pi^2$ (C) $22\pi^2$ (D) $18\pi^2$

Ans. C

Sol. $16(\sec^{-1}x)^2 + (\operatorname{cosec}^{-1}x)^2$

$$\sec^{-1} x = a \in [0, \pi] - \left\{ \frac{\pi}{2} \right\}$$

$$\operatorname{cosec}^{-1} x = \frac{\pi}{2} - a$$

$$= 16 \left[a^2 + \left(\frac{\pi}{2} - a \right)^2 \right] = 16 \left[2a^2 - \pi a + \frac{\pi^2}{4} \right]$$

$$\max]_{a=\pi} = 16 \left[2\pi^2 - \pi^2 + \pi \frac{2}{4} \right] = 20\pi^2$$

$$\min]_{a=\frac{\pi}{4}} = 16 \left[\frac{2 \times \pi^2}{16} - \frac{\pi^2}{4} + \frac{\pi^2}{4} \right] = 2\pi^2$$

$$\text{Sum} = 22\pi^2$$

75. If $\frac{\pi}{2} \leq x \leq \frac{3\pi}{4}$, then $\cos^{-1} \left(\frac{12}{13} \cos x + \frac{5}{13} \sin x \right)$ is equal to :

यदि $\frac{\pi}{2} \leq x \leq \frac{3\pi}{4}$ है, तो $\cos^{-1} \left(\frac{12}{13} \cos x + \frac{5}{13} \sin x \right)$ बराबर है :

- (A) $x - \tan^{-1} \frac{5}{12}$ (B) $x + \tan^{-1} \frac{4}{5}$ (C) $x + \tan^{-1} \frac{5}{12}$ (D) $x - \tan^{-1} \frac{4}{3}$

Ans. A

Sol. $\cos^{-1}(\cos \alpha \cos a + \sin \alpha \sin x)$ $\cos \alpha = \frac{12}{13}$

$$\cos^{-1}(\cos(x - \alpha))$$

$$= x - \alpha$$

$$= x - \tan^{-1}\left(\frac{5}{12}\right)$$

$$\sin \alpha = \frac{5}{13}$$

$$\tan \alpha = \frac{5}{12}$$

76. If $\alpha > \beta > \gamma > 0$, then the expression $\cot^{-1}\left\{\beta + \frac{(1+\beta^2)}{(\alpha-\beta)}\right\} + \cot^{-1}\left\{\gamma + \frac{(1+\gamma^2)}{(\beta-\gamma)}\right\} + \cot^{-1}\left\{\alpha + \frac{(1+\alpha^2)}{(\gamma-\alpha)}\right\}$ is equal to :

यदि $\alpha > \beta > \gamma > 0$ है, तो व्यंजक $\cot^{-1}\left\{\beta + \frac{(1+\beta^2)}{(\alpha-\beta)}\right\} + \cot^{-1}\left\{\gamma + \frac{(1+\gamma^2)}{(\beta-\gamma)}\right\} + \cot^{-1}\left\{\alpha + \frac{(1+\alpha^2)}{(\gamma-\alpha)}\right\}$ बराबर है :

- (A) 0 (B) π (C) $\frac{\pi}{2} - (\alpha + \beta + \gamma)$ (D) 3π

Ans. B

Sol. $\cot^{-1}\left(\frac{\alpha\beta+1}{\alpha+\beta}\right) + \cot^{-1}\left(\frac{\beta\gamma+1}{\beta+\gamma}\right) + \cot^{-1}\left(\frac{\gamma\alpha+1}{\gamma+\alpha}\right)$

$$\Rightarrow \tan^{-1}\left(\frac{\alpha-\beta}{1+\alpha\beta}\right) + \tan^{-1}\left(\frac{\beta-\gamma}{1+\beta\gamma}\right) + \left(\pi + \tan^{-1}\left(\frac{\gamma-\alpha}{1+\gamma\alpha}\right)\right)$$

$$\Rightarrow \tan^{-1}(\alpha) - \tan^{-1}\beta + \tan^{-1}\beta - \tan^{-1}\gamma + \pi + \tan^{-1}\gamma - \tan^{-1}\alpha = \pi$$

77. Given that the inverse trigonometric function assumes principal values only. Let x, y be any two real numbers in $[-1, 1]$ such that $\cos^{-1}x - \sin^{-1}y = \alpha$, $-\frac{\pi}{2} \leq \alpha \leq \pi$. Then the minimum value of $x^2 + y^2 + 2xy \sin \alpha$ is :

दिया है कि प्रतिलोम त्रिकोणमिति फलन केवल मुख्य मान मानता है। माना $[-1, 1]$ में दो वास्तविक संख्याएँ x तथा y इस प्रकार हैं कि

$\cos^{-1}x - \sin^{-1}y = \alpha$, $-\frac{\pi}{2} \leq \alpha \leq \pi$ तो $x^2 + y^2 + 2xy \sin \alpha$ का न्यूनतम मान है :

- (A) -1 (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) 0

Ans. D

Sol. $\cos^{-1}x - \left(\frac{\pi}{2} - \cos^{-1}y\right) = \alpha$

$$\cos^{-1} x + \cos^{-1} y = \frac{\pi}{2} + \alpha$$

$$\alpha \in \left[-\frac{\pi}{2}, \pi\right], \frac{\pi}{2} + \alpha \in \left[0, \frac{3\pi}{2}\right]$$

$$\cos^{-1} \left(xy - \sqrt{1-x^2} \sqrt{1-y^2} \right) = \frac{\pi}{2} + \alpha$$

$$xy - \sqrt{1-x^2} \sqrt{1-y^2} = -\sin \alpha$$

$$(xy + \sin \alpha)^2 = (1-x^2)(1-y^2)$$

$$x^2 y^2 + 2xy \sin \alpha + \sin^2 \alpha = 1 - x^2 - y^2 + x^2 y^2$$

$$x^2 + y^2 + 2xy \sin \alpha = 1 - \sin^2 \alpha$$

$$x^2 + y^2 + 2xy \sin \alpha = \cos^2 \alpha$$

$$\text{Minimum value of } \cos^2 \alpha = 0$$

$$\text{At } \alpha = \frac{\pi}{2}$$

78. Let $x = \frac{m}{n}$ (m, n are co-prime natural numbers) be a solution of the equation $\cos(2 \sin^{-1} x) = \frac{1}{9}$ and let α, β ($\alpha > \beta$) be the roots of the equation $mx^2 - nx - m + n = 0$. Then the point (α, β) lies on the line :

माना समीकरण $\cos(2 \sin^{-1} x) = \frac{1}{9}$ का एक हल $x = \frac{m}{n}$ (m, n असहभाज्य घनपूर्णांक हैं) है और माना समीकरण

$mx^2 - nx - m + n = 0$ के मूल α, β ($\alpha > \beta$) हैं। तो बिन्दु (α, β) किस रेखा पर है :

- (A) $5x + 8y = 9$ (B) $5x - 8y = -9$ (C) $3x + 2y = 2$ (D) $3x - 2y = -2$

Ans. A

Sol. $\sin^{-1} x = \theta \in I^{\text{st}} \text{ quadrant}$

$$\cos 2\theta = \frac{1}{9}$$

$$\Rightarrow 1 - 2\sin^2 \theta = \frac{1}{9}$$

$$\Rightarrow \sin^2 \theta = \frac{4}{9}$$

$$\Rightarrow x^2 = \frac{4}{9} \Rightarrow x = \frac{2}{3} = \frac{m}{n}$$

$$\text{So } mx^2 - nx - m + n = 0$$

$$\Rightarrow 2x^2 - 3x + 1 = 0$$

roots are $1, \frac{1}{2}$

$$\alpha = 1, \beta = \frac{1}{2}$$

So $(\alpha, \beta) = \left(1, \frac{1}{2}\right)$ lies on $5x + 8y = 9$

79. For $\alpha, \beta, \gamma \neq 0$, if $\sin^{-1}\alpha + \sin^{-1}\beta + \sin^{-1}\gamma = \pi$ and $(\alpha + \beta + \gamma)(\alpha - \gamma + \beta) = 3\alpha\beta$, then γ equals :

$\alpha, \beta, \gamma \neq 0$, के लिए, यदि $\sin^{-1}\alpha + \sin^{-1}\beta + \sin^{-1}\gamma = \pi$ है तथा $(\alpha + \beta + \gamma)(\alpha - \gamma + \beta) = 3\alpha\beta$ है, तो γ बराबर है :

- (A) $\frac{1}{\sqrt{2}}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\sqrt{3}$ (D) $\frac{\sqrt{3}-1}{2\sqrt{2}}$

Ans. B

Sol. Let $\sin^{-1}\alpha = A$, $\sin^{-1}\beta = B$ and $\sin^{-1}\gamma = C$

$$\therefore A + B + C = \pi$$

$$\text{Also, } (\alpha + \beta + \gamma)(\alpha - \gamma + \beta) = 3\alpha\beta$$

$$\Rightarrow (\alpha + \beta)^2 - \gamma^2 = 3\alpha\beta$$

$$\Rightarrow \alpha^2 + \beta^2 - \gamma^2 = \alpha\beta$$

$$\Rightarrow \frac{\alpha^2 + \beta^2 - \gamma^2}{2\alpha\beta} = \frac{1}{2}$$

$$\Rightarrow \cos C = \frac{1}{2} \Rightarrow C = 60^\circ$$

$$\therefore \sin C = \frac{\sqrt{3}}{2} = \gamma$$

80. If $a = \sin^{-1}(\sin(5))$ and $b = \cos^{-1}(\cos(5))$, then $a^2 + b^2$ is equal to :

यदि $a = \sin^{-1}(\sin(5))$ तथा $b = \cos^{-1}(\cos(5))$, तो $a^2 + b^2$ बराबर है :

- (A) $4\pi^2 + 25$ (B) $4\pi^2 - 20\pi + 50$ (C) $8\pi^2 - 40\pi + 50$ (D) 25

Ans. C

Sol. $a = \sin^{-1}(\sin 5) = 5 - 2\pi$

$$\text{and } b = \cos^{-1}(\cos 5) = 2\pi - 5$$

$$\therefore a^2 + b^2 = (5 - 2\pi)^2 + (2\pi - 5)^2$$

$$= 8\pi^2 - 40\pi + 50$$